Jeffrey Fan.

I make apps, art, robots, and more.

An aspiring creator looking to build things with the fundamentals of biology, chemistry, physics, and computers. Ready to make more tools for more people.

in

EDUCATION

2022 – 2026
Palo Alto High School

- GPA: 4.0 / 4.0

SKILLS

Programming

- Python
- Numpy
- SciPy
- Tkinter
- PyGame
- JS
- TS
- HTML
- CSS
- NodeJS
- ElectronJS
- ThreeJS
- Redux
- React
- RTKQ
- Tailwind
- Java
- OOP
- GLSL

Physics

- Newtonian (Calculus)
- Lagrangian

General Biology

General Chemistry

Graphic Design

WORK EXPERIENCE

➢ **A0 Systems – Hopper Division** — Jun – Aug 2025

Paid Intern

- Worked on front-end with React, Redux, and Tailwind
- Refactored and improved the robustness of the entire UI framework, fixed countless bugs and inefficiencies
- Average PR had less than -500 in line count with functional and maintenance improvements
- Worked to enhance AI code generation via facilitated frameworks, created numerous features for the application entirely solo (video recording and analysis, chats/tooling interactions, generic agent framework)

➢ **Lawrence Berkeley National Laboratory** — Feb 2024 – 2025

Paid Student Research Assistant

- Operated on a large scale with supercomputing to evaluate mathematical optimization tasks
- Performed geometric optimization and frequency analysis of molecules to determine reaction efficiency
- Worked on a succinct CL tool for batch submitting and tracking repetitive tasks to Slurm
- Performed statistical analysis of DNA structures using alternate coordinate representations to determine their forms

RESEARCH EXPERIENCE

➢ **Student Research Assistant** @ Stanford — Aug 2023 – 2025

- Participated in a large research team
- Studied relationship between protein buildups and restrictive cardiomyopathy. Focused on echocardiogram image analysis and processing, along with data processing with Python
- Paper is currently under review

➢ **IsPETase Optimization** @ LBNL — Jan – Oct 2025

- Ran molecular geometry optimization and vibrational analysis to determine reaction energies and favorability
- Discovered and proved certain enzymatic-enhancing mutations
- Analyzed existing literature to provide new research directions and future biochemical experimental confirmation

AWARDS

- FRC Autonomous Award
- FRC Innovation in Control Award
- USACO Silver
- AIME Qualifier
- Congressional App Challenge Honorable
- FIC Honorable
- SVTT Honorable
- Paly Science Award
  - Biology H
  - Chemistry H
  - AP Physics C
- Congressional Art Competition Honorable
- Scholastic Art Gold Key
- Scholastic Art Silver Key
- PVSA

LANGUAGES

English – Fluent
Spanish – AP
Chinese – Fluent

LEADERSHIP

- Software Captain @ FRC
- Group Lead @ COSMOS
- CTO and Group Lead @ Noteblast, ACF
- Lab Group Lead @ SSSIP

EXTRACURRICULARS

➤ FIRST Robotics Competition — 2023 – 2026

Software Captain of 6036

- Management of a team of over 30 software recruits
- Robot competes in 3 annual regionals
- 2026: Top 100 of 3700 teams in the world, top 10 of 290 in California.
- Created a robust robot development environment
  - Necessary debugging and tuning tools built in for rapid testing in the short 2-month season
  - Modularity and collaboration-focused software architecture promotes sustainability
  - Versatile custom library speeds up development tenfold
- Created a front-end application for diagnostic interaction with the robot during a competition
- Developed an entire scouting system for information gathering during FIRST competitions
- Created a fully custom vision system software interacting with custom hardware for x8 performance improvement of AprilTag detection compared to open source solutions

➤ COSMOS UC Davis – Computational Biology — July 2024

Group Lead

- Led a group of 4 through modeling numerous biological systems
- Created a front-end application capable of simulating numerous mathematical models
  - Fitz-Nagumo neuron and heart action potential model
  - Gierer-Meinhardt animal coat pattern formation model
  - Boid and Viseck flocking model
- Presented an analysis on Gierer-Meinhardt pattern amplification frequency related to initial empirical model parameters to professors
- Wrote a write-up of our discoveries and learning process

➤ Applied Computing Foundation (ACF) – NoteBlast — 2022 – 2024

CTO and Group Lead

- Solely created a full-stack application called Noteblast, providing musicians with free instrument tuning and built-in sightreading gameplay
- Congressional App Challenge Honorable mention
- SVTTT 9th Summit participant
- FIC Honorable mention
- Author of a published paper about music and education sustainability in JSR

➤ Stanford Summer Science Internship Program (SSSIP) — July 2023

Lab Group Lead

- Led a lab group of 4 through experiments and data gathering
- Organized and designed the final presentation for Stanford professors
- Developed an image processing algorithm (and later an app) to count transfected cells, which surprised camp mentors

PROJECTS

- **Refraction**

  A ray optics simulator with refraction, reflection, material customization, spectrogram customization, and more! Still a work in progress.

- **Reaction TrajFinder**

  An ORCA software output file parser to understand reaction dynamics and energy wells through data processing.

- **LuxCM5**

  Custom software for custom circuit cameras. Features UMich AprilTag detector optimizations and full frontend application. Beats open source solutions by 8x FPS.

- **Visual Art Gallery V2**

  My second iteration of my virtual art gallery, with a strong focus on light and immersion. I really enjoyed the super dynamic camera effect I made. Check it out!

- **CellLuminex**

  A fluorescent cell counter and calculator, able to distinguish immunofluorescently-stained cell images.

- **Portfolio V3**

  My third iteration of my portfolio website. Built using a variety of frameworks, including React, Vite, TailwindCSS, and Framer-Motion.

- **NoteBlast**

  A sight-reading rhythm game to help musicians practice! Comes with a built-in free tuner to provide feedback on the accuracy of your instrument

- **Merge Game**

  A fun 2048-like Chrome extension about merging shapes to create higher tiers. A good offline way to enjoy some downtime. Boasting over 3.6K installations in total and over 700 monthly users.

- **Biotech Sandboxes**

  A sandbox environment where I simulated some mathematical models of biological systems! Notable models include Gierer-Meinhardt and FitzHugh-Nagumo.

- **pPatrol**

  2024 FRC Crescendo web scouting application linked with PythonAnywhere simple Flask backend.

- **PeninsulaPortal**

  A complete FRC robot diagnostic and debugger, path generator, and competition pit display. Sleek, navigable design focused on user experience. Full end-to-end development from back-end servers to front-end applications.

- **Celestial.js**

  A top-down space shooter game where you defeat asteroids and enemies. Includes in-house self-developed JavaScript game engine.

VOLUNTEERING

➤ **ACME Education Group** – *150+ hr, 4 mo*

- Volunteered at a local education group during the summer
- Provided teaching aid and general guidance for elementary schoolers
- Became a favorite among the kids

➤ **Peninsula Academy: Build-a-Bot** – *150 hr, 3 wk*

- Volunteered at robotics team summer outreach program as counselor lead
- Guided middle schoolers towards building a fully functioning FRC robot
- Taught software and Java basics, and basic motion control theory

➤ **Peninsula Academy: Fall Classes** – *50 hr, 10 wk*

- Volunteered at robotics team fall classes as sole programming teacher
- Designed a comprehensive curriculum from zero programming experience to a fluent Python programmer
- Of the three fall classes offered, this course received no negative reviews

PUBLICATIONS

1. **Fan J**, Ha Y, Active Site Local Environment Allows Acidic and Basic Synergy in Enzymatic Ester Hydrolysis by PETase, ***Computational and Structural Biotechnology Journal***, Under review

2. Wang W, Liu Z, Zheng L, Shi Y, Manzo-Casio E, Shi S, **Fan J**, Zhou D, Wang W, Zhang X, Yang Y, Too rigid to strike: Excess desmin at Z-discs underlies a restrictive cardiomyopathy in GAN mice, ***Circulation Research***, Under review

3. **Fan J**, Ha Y. Micro- and Nanoplastics and the Immune System: Mechanistic Insights and Future Directions. ***Immuno***. 2025; 5(4):52. https://doi.org/10.3390/immuno5040052

4. Rong Z, Hong LG, Huo YY, Li J, Zheng DQ, Ha Y, **Fan J**, Xu XW, Wu YH. Molecular Insight Into the Hydrolysis of Phthalate Esters by a Family IV Esterase. ***Environ Microbiol***. 2025 Jul;27(7):e70134. doi: 10.1111/1462-2920.70134. PMID: 40600832.

5. Kang A*, **Fan J***, Battulga A, & Choi H. Addressing the Issue of Musical Education through Gamified Tuning Analysis. ***Journal of Student Research*** 2024, 13(1). https://doi.org/10.47611/jsrhs.v13i1.6235 * Equal contribution

6. Mai G, **Fan J**, Mai B, & Fan X. Effect of Music Therapy on Alzheimer’s Disease: How Music Combats Alzheimer’s Disease?. ***International Journal of Innovative Research in Medical Science***. 2023, 8(07), 279–286. https://doi.org/10.23958/ijirms/vol08-i07/1702

PATENTS

- Jeffrey Fan, Yang Ha. 2025. "Engineered PETase Variants and Methods for Enhancing Ester Hydrolysis via Acidic-Basic Synergistic Local Environments." U.S. Patent 63/913,260, Provisional.

BOOKS

- Hand-drawn comic book series (3 parts)
- Personal art showcase book collection (2 parts)